

Daniel Gray

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EDUCATION

California State University, East Bay

Hayward, CA

B.S. Computer Science Graduated December 2025

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Artificial Intelligence, Machine Learning, Operating Systems

TECHNICAL SKILLS

Languages: Java, Python, C++

Backend & Cloud: Spring Boot, REST APIs, PostgreSQL, Redis, Docker, AWS, Git, Linux

Practices: Agile, SDLC, OOP, Layered Architecture, Unit Testing, Code Review

Data & ML: PyTorch, OpenCV, Natural Language Processing, LLMs, Agents

EXPERIENCE

PoseTek | FullStack Software Engineer Intern (Start up)

Pleasanton CA April 2026 – Present

- Designed the backend infrastructure for PoseTek's pose analysis platform using Python and FastAPI, deployed on Railway, with Supabase handling authentication and the database layer.
- Architected a lightweight data pipeline where joint keypoints are extracted frame by frame on device (Android/Kotlin, iOS/Swift) and transmitted as JSON payloads to the backend, eliminating video uploads and significantly reducing latency for real time feedback

Cal State East Bay | AI Researcher

Hayward, CA Sep 2024 – Aug 2025

- Trained and iterated on custom YOLO and PyTorch models for object detection and classification, validating performance against labeled underwater video datasets and tuning hyperparameters to improve accuracy across multiple training cycles
- Developed end to end ML pipelines covering data preprocessing, model training, evaluation, and inference, deploying computer vision models on embedded hardware for live video classification tasks

UL Solutions | Wireless Testing Intern

Fremont, CA May 2021 – Aug 2022

- Achieved zero defects across all certification reports submitted to regulators by executing Bluetooth and Wi-Fi compliance testing against global standards, through systematic test procedures and meticulous documentation.
- Reduced new intern onboarding time by proactively creating and delivering training sessions on RF emissions testing methodologies, enabling junior team members to run tests independently within their first two weeks.

PROJECTS

Chess Insights API

Java, Spring Boot, PostgreSQL, Redis, React, Docker

- Built a full stack chess analytics platform that ingested 400+ games from the Chess.com public API into PostgreSQL, computing performance insights across 198 tracked openings and surfacing metrics like win rate and peak rating through custom REST endpoints.
- Engineered analytics endpoints using custom JPQL and native SQL aggregation queries that compute win rate by color, average opponent strength, peak/average rating, and recommended openings, all filterable by time range.
- Reduced API response times by 90% (from ~430ms to ~50ms) for frequently requested analytics by implementing Redis caching with a 1 hour TTL and automatic cache invalidation on data sync.
- Automated daily data freshness by building a scheduled nightly sync job with Spring @Scheduled, backed by a SyncJob tracking table that logs fetch counts, status, and errors for full pipeline observability
- Built a repertoire training system with PGN parsing, spaced repetition drill mode, and deviation analysis, enabling users to build, save, and practice custom opening lines against their actual game history.
- Shipped a React frontend dashboard with live stats visualization, rating progression charts, and win rate by color breakdowns, connected to the backend via JWT authentication.

BlueROV Autonomous Underwater Robot

Python, PyTorch, OpenCV, ROS, Pymavlink

- Integrated YOLO object detection with OpenCV and PyTorch into a real time computer vision pipeline running on a submersible robot, processing live HD video frames to detect and classify marine species with low latency inference aboard the BlueROV platform
- Built a closed loop perception to action pipeline that fed YOLO detection outputs directly into PyMAVLink control commands, enabling autonomous underwater tracking by translating real time bounding box data into thruster actuation signals